CrossRoute: Research Brief, Technical Requirements & Presentation Dossier

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CrossRoute: Research Brief, Technical Requirements & Presentation Dossier

Cyber-Physical Zero-Genset Cold Corridor for Port Drayage & Yulon Green Mobility

Project Origin: 2026 Sustainable Smart Innovation Hackathon (2026 永續智慧創新黑客松)
Challenge Track: Track 15-M "AI Empowering the Future: Sustainable Solutions for Enterprises & Society"
Lead Sponsoring Enterprise: Yulon Group (裕隆集團) in coordination with Taiwan International Ports Corporation (TIPC)
Academic Partner: National Kaohsiung Normal University (NKNU, 國立高雄師範大學)
Live Interactive Digital Twin: https://imteyaz087.github.io/crossroute-port/

1. Executive Summary & Context

This briefing dossier is compiled for engineering researchers, team members, and independent evaluators investigating the **CrossRoute** cold-chain telematics architecture.

CrossRoute was created in response to a specific structural problem in maritime port logistics: **how to eliminate the 1,200 kg clip-on diesel generator (Genset) used during short-haul refrigerated container drayage without compromising food safety, and how to enable commercial electric vehicles (such as Yulon's CMC-ET35) to enter the refrigerated logistics market without deadweight battery penalties.**

2. The Exact Problem the Organizers Expect Solved

2.1 The Industry Bottleneck: The 1,200 kg "Clip-On Genset" Trap

Taiwan's Port of Kaohsiung handles over **400,000 TEU** of refrigerated (reefer) containers annually across Terminal 4 (Berths 63—66) and Terminal 5 (Berths 77—81). These containers carry high-value, temperature-sensitive cargo: imported frozen meats (-18°C), sashimi tuna (-55°C), pharmaceuticals and vaccines (2°C to 8°C), and premium export fruit.

When an ocean container is discharged from a container ship and loaded onto a drayage truck chassis for the short **35 to 55-minute haul** across the Kaohsiung Cross-Harbor Tunnel to the Qianzhen Bonded Logistics Center or Xiaogang cold storage hubs:

1. **The Tractor Cannot Power the Container**: Standard semi-tractors lack the high-voltage 460V 3-phase electrical supply required by marine reefer units (Carrier Transicold, Daikin, Thermo King).
2. **The Default Industry "Workaround"**: Logistics operators clamp a standalone **1,200 kg diesel generator ("Genset")** onto the container chassis directly behind the truck driver's cabin.

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Traditional Short-Haul Port Drayage:
[ Diesel Tractor ] ── [ 1,200 kg Clip-on Genset ] ── [ 40ft High-Cube Reefer ]
                      ↳ 85 dB Noise, Toxic PM2.5, NT$ 1,610 to 1,900 Cost per Trip
```

2.2 The Quadruple Damage of the Status Quo

1. **Severe Operating Expense (OPEX)**:
   - Genset rental fees: **NT$ 1,500 / day** (approx. NT$ 750 / trip).
   - Genset diesel consumption: **3.2 Litres / hour** (approx. NT$ 260 / trip).
   - Extra diesel burned by the tractor towing 1.2 tons of deadweight: approx. **NT$ 100 / trip**.
   - Net operational penalty: **NT$ 1,110 to NT$ 1,610 per container trip**.
2. **Driver Health & Occupational Injury**:
   - The Genset operates 50 cm behind the driver's head, emitting continuous **85 dB low-frequency vibration** and concentrated $\mathrm{PM}_{2.5}$ and NO_x carcinogenic exhaust throughout traffic delays.
3. **The Electric Vehicle Blockade**:
   - Taiwan's commercial EV tractors (such as Yulon's **China Motor Corporation CMC-ET35**) cannot bear an additional 1,200 kg deadweight generator without suffering catastrophic battery drain and payload capacity loss.
4. **Demurrage & Gate Congestion**:
   - Drayage trucks idle for an average of **48 minutes** at terminal gates due to disconnected terminal operating schedules, risking **NT$ 4,500 to NT$ 9,000 / day** in shipping line demurrage penalties.
5. **Scope 3 Carbon Audit Gap**:
   - Global exporters facing the European Union's Carbon Border Adjustment Mechanism (CBAM) or domestic carbon fees lack verified, empirical Well-to-Wheel (WTW) cold-chain emissions data conforming to **ISO 14083**.

3. What Deliverables the Organizers & Judges Expect

The hackathon committee (Yulon Group executives, automotive engineers, and port authorities) evaluates entries against five explicit deliverables:

Deliverable	Organizers' Core Question	CrossRoute Engineering Solution
1. Scientific Proof	"How can you guarantee food safety if you cut off generator power during transit?"	**Dynamic Thermal Budget Engine (DTBE)** based on transient Fourier heat conduction and cargo thermal mass inertia.
2. Non-Invasive Hardware	"How do you connect to trucks and reefers without voiding vehicle and container warranties?"	**ISO 11898-2 opto-isolated passive CAN tap** reading SAE J1939 broadcast frames (PGN 61444 / 65269) with zero signal injection.

Deliverable	Organizers' Core Question	CrossRoute Engineering Solution
3. Yulon Synergy	"How does this integrate 2 to 3 distinct Yulon Group business units?"	Tri–party architecture: **CMC** (ET35 EV trucks) + **Car Plus** (FMS fleet dispatch) + **TAC** (green leasing & carbon financing).
4. CFO Unit Economics	"What are the real–world dollar savings and carbon abatement numbers?"	Detailed unit economics: **NT$ 1,610 saved per trip**, 14.2–month fleet payback, and **−77.7% Scope 3 CO_2e reduction**.
5. Working Demonstrator	"Do you have a working, interactive prototype juries can evaluate?"	Full PBR digital twin with live thermodynamic physics deployed at `https://imteyaz087.github.io/crossroute-port/`.

4. The Core Scientific Breakthrough: The "Virtual Thermal Battery"

4.1 First–Principles Thermodynamic Derivation

The foundational insight of CrossRoute is that logistics operators mistakenly treat a 40–foot container as an empty box that rapidly loses temperature, ignoring the massive thermal inertia of frozen cargo:

1. **Reefer Insulation Rating ($U \cdot A$)**: A standard 40ft High–Cube container insulated with 50 mm polyurethane foam has an overall heat transmission coefficient:
$$U \cdot A \approx 45 \text{ W/°C} \quad (0.38 \text{ W/(m}^2 \cdot \text{K) across } 142.8 \text{m}^2)$$
2. **Extreme Ambient Temperature Gradient (ΔT)**: In Kaohsiung summer heat ($T_{amb} = 32°C$), with deep–frozen cargo at $T_{box} = -18°C$:
$$\Delta T = 32 - (-18) = 50°C$$
3. **Heat Ingress Rate ($\dot{Q}_{ingress}$)**:
$$\dot{Q} = 45 \text{ W/°C} \times 50°C = 2,250 \text{ W} = 2.25 \text{ kW}$$
4. **Total Heat Ingress During a 38–Minute Transit (Q_{total})**:
$$Q_{total} = 2.25 \text{ kW} \times \left(\frac{38}{60} \text{ hr}\right) = 1.425 \text{ kWh} \approx 5.13 \text{ MJ}$$
5. **Thermal Mass Capacity of 20 Tons of Frozen Cargo (C_{cargo})**: With specific heat capacity $c \approx 1,800 \text{ J/(kg} \cdot \text{°C)}$:
$$C_{cargo} = m \cdot c = 20,000 \text{ kg} \times 1,800 \text{ J/(kg} \cdot \text{°C)} = 36,000,000 \text{ J/°C} = 36 \text{ MJ/°C}$$
6. **Core Temperature Drift During Unpowered Drayage (ΔT_{cargo})**:
$$\Delta T_{cargo} = \frac{Q_{total}}{C_{cargo}} = \frac{5.13 \text{ MJ}}{36 \text{ MJ/°C}} = \mathbf{+0.14°C}$$

Regulatory Tolerance: International HACCP and Taiwan TFDA regulations mandate that frozen food core temperatures remain below –18°C, with a maximum allowable transit deviation of $\mathbf{\pm 1.5°C}$.
Result: An unpowered 38–minute transit experiences a temperature rise of $\mathbf{+0.14°C}$—leaving a safety buffer **10 times wider than required by law**.

5. Comparison: Existing Refrigeration & Energy–Saving Methods

When researching existing commercial solutions, evaluate them against CrossRoute:

Method	Capital Expenditure (CAPEX)	Weight Penalty	Operational Complexity	Suitability for Short Port Drayage
Traditional Diesel Genset	Low rental (NT$1,500/day)	Heavy (+1,200 kg)	Fueling, maintenance, noise, toxic exhaust	Current status quo; environmentally unacceptable.
Battery Electric Genset (e.g. Carrier Vector eCool)	Very High (>NT$ 800,000 / unit)	Heavy (+800 kg battery pack)	400V high-voltage charging, battery degradation	Overkill for 40-min drayage; prohibitive fleet CAPEX.
Eutectic Cold Plates	High (NT$ 300,000 / truck)	Very Heavy (+1,500 kg beams)	Requires overnight freezing; fixed temperature	Inflexible; incompatible with standard marine containers.
CrossRoute Dynamic Thermal Budgeting (DTBE)	**Zero added hardware cost** (SaaS + NT$8.5k CAN tap)	**Zero deadweight** (1.2T eliminated)	Pure software telematics & pre-chill coordination	**Optimal**: 85% cost savings, zero emissions, instant EV compatibility.

6. Official Competition Pitch Script (English & Traditional Chinese)

This script is structured for a 3-minute jury presentation:

Part 1: The Hook (0:00 — 0:45)

- **English**:
  "Every day, over 1,000 refrigerated containers arrive at the Port of Kaohsiung carrying frozen meats, vaccines, and seafood. To haul them just 15 kilometers across the harbor tunnel to the logistics center, trucking fleets clamp a heavy, 1,200 kg diesel generator onto every single truck. It burns diesel, creates 85 decibels of toxic noise right behind the driver's head, and costs NT$ 1,600 per trip. Worst of all, this 1.2-ton deadweight makes it impossible for electric trucks to enter the market. But why are we burning diesel to cool cargo that is already frozen solid?"
- **Traditional Chinese (中文)**:
  「每天有超過一千只冷凍貨櫃抵達高雄港。為了短短 15 公里的過港隧道短程拖運，每輛拖車都在司機後腦勺外掛一具重達 1,200 公斤的柴油發電機。它每趟燒掉近四百元柴油、發出 85 分貝致癌噪音，每趟虛耗 1,600 元。更致命的是，這 1.2 噸的死重讓商用電動卡車根本無法切入冷鏈市場。但我們為什麼要燒柴油，去冷卻已經結成冰塊的貨物？」

Part 2: The Breakthrough (0:45 — 1:30)

- **English**:
  "CrossRoute redefines refrigerated cargo from a cooling burden into a virtual thermal battery. A 20-ton frozen container contains 36 million Joules per degree of thermal inertia. Through Fourier heat conduction calculations, we proved that during a 45-minute haul, the core temperature rises by only 0.14°C to 0.41°C—well within the international HACCP safety threshold. By tapping into vehicle CAN-bus data passively via opto-isolated hardware, CrossRoute calculates the exact dynamic thermal budget in real time. If severe tunnel traffic occurs, our dual-fail-safe protocol routes the truck to backup charging points before any quality loss can happen."
- **Traditional Chinese (中文)**:
  「CrossRoute 顛覆全球冷鏈思維：20 噸的冷凍貨物本體就是一顆蘊含 36 MJ/°C 的巨型冷能蓄熱池！透過三維

傅立葉熱傳導運算，我們證實拔掉發電機後，45 分鐘路程中核心溫度僅上升 0.14°C，完全在國際食安規範允許範圍內。透過光電隔離線束被動監聽車載 CAN-bus，CrossRoute 動態熱預算系統秒級監控冷能餘裕；若遇過港隧道嚴重事故，自動引導至備援路徑或中繼充電樁補冷，確保百分之百零貨損。」

Part 3: The Yulon Ecosystem Synergy (1:30 – 2:15)

- **English**:
  "CrossRoute unifies three of Yulon Group's flagship business units into one green logistics loop: First, China Motor Corporation (CMC) provides ET35 commercial electric trucks that can now enter cold-chain drayage with zero deadweight penalty, saving 22% battery range. Second, Car Plus manages the smart fleet dispatching network using our digital pass. Third, Taiwan Acceptance Corporation (TAC) finances the fleets through green leases, where verified ISO 14083 carbon credits directly offset leasing interest rates."
- **Traditional Chinese (中文)**:
  「CrossRoute 深度整合裕隆集團三大事業版圖：第一，中華汽車 (CMC) ET35 商用電動車無負擔切入冷鏈市場，利用熱預算釋放 22% 電池續航；第二，格上租車 (Car Plus) 導入智慧車聯網調度派單；第三，裕融企業 (TAC) 推出冷鏈永續租賃金融，將符合 ISO 14083 的實測減碳憑證轉化為融資利息抵減，打造綠色移動閉環。」

Part 4: The Bottom Line (2:15 – 3:00)

- **English**:
  "The financial numbers are undeniable: CrossRoute saves NT$ 1,610 per container trip, saving a 30-truck fleet over NT$ 1.5 million every single month with a payback period under 15 months. Environmentally, it cuts transport carbon emissions by 78%, eliminating 9,600 tons of CO_2 and 28 tons of PM2.5 annually in Kaohsiung. Our working digital twin is live at crossroute-port on GitHub Pages. CrossRoute isn't just an idea—it is the turnkey blueprint for Taiwan's zero-emission port corridors."
- **Traditional Chinese (中文)**:
  「財務效益真金白銀：CrossRoute 每趟實質淨省 NT$ 1,610 元，三十輛車隊每月現省超過 150 萬元，投資回收期不到 15 個月；環境效益每年為高雄市減少 9,600 噸碳排放與 28 噸 PM2.5 致癌煙塵。我們完整的數位孿生平台已於 GitHub Pages 公開上線。CrossRoute 不僅是一個構想，而是台灣港口邁向淨零冷鏈的最佳解方！」

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7. Key Research Questions for Your Investigation

To help you conduct independent, rigorous technical research:

1. **Heat Transmission Coefficient (U-value)**: Verify the thermal conductivity ($k \approx 0.024\ \mathrm{W/(m \cdot K)}$) and insulation degradation rate of marine reefer container polyurethane walls over 5 to 10 years of service.
2. **Latent Heat vs. Sensible Heat**: Examine the phase-change enthalpy buffer in pre-chilled meat/seafood at temperatures between –20°C and –18°C.
3. **SAE J1939 CAN-bus Standard**: Review PGN 61444 (Electronic Engine Controller 1) and PGN 65269 (Ambient Conditions) to confirm passive broadcast rates without active request querying.
4. **ISO 14083:2023 Rules**: Review Section 7 on energy allocation rules between vehicle traction energy and auxiliary cargo cooling energy in intermodal freight chains.